REPRESENTABILITY OF A TRANSFORMATION DETECTED ON NOETHERIAN AFFINES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4886-4903 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 4905-4993, the complete author proof (87 body lines).

Editorial note. Navigation stubs: 10 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 4 of queue_artin.json (pb=4309, stmt 4886-4903, proof 4905-4993).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00QN: sources/stacks-artin/pinned-original-source.tex, source lines 4993–4993. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 001X: sources/stacks-artin/pinned-original-source.tex, source lines 4993–4993. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01LH: sources/stacks-artin/pinned-original-source.tex, source lines 4993–4993. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GE3: sources/stacks-artin/pinned-original-source.tex, source lines 4830–4840. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GE4: sources/stacks-artin/pinned-original-source.tex, source lines 4858–4862. Cited by the retained statement or proof but not part of

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the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZC: sources/stacks-artin/pinned-original-source.tex, source lines 4993–4993. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05LX: sources/stacks-artin/pinned-original-source.tex, source lines 4993–4993. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 049M: sources/stacks-artin/pinned-original-source.tex, source lines 4993–4993. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GDZ: sources/stacks-artin/pinned-original-source.tex, source lines 4993–4993. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GDY: sources/stacks-artin/pinned-original-source.tex, source lines 4993–4993. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$. Let $F', G' : (Sch/S)^{opp}_\tau \to Sets$ be limit preserving and sheaves. Let $a' : F' \to G'$ be a transformation of functors. Denote $a : F \to G$ the restriction of $a' : F' \to G'$ to $(Noetherian/S)_\tau$. The following are equivalent*

1. *a' is representable (as a transformation of functors, see Categories, Definition 001X), and*
2. *for every object V of $(Noetherian/S)_\tau$ and every map $V \to G$ the fibre product $F \times_G V : (Noetherian/S)^{opp}_\tau \to Sets$ is a representable functor, and*
3. *same as in (2) but only for V affine finite type over S mapping into an affine open of S.*

Proof. Assume (1). By Limits of Spaces, Lemma 0GDY the transformation a' is limit preserving[1]. Take $\xi : V \to G$ as in (2). Denote $V' = V$ but viewed as an object of $(Sch/S)_\tau$. Since G is the restriction of G' to $(Noetherian/S)_\tau$ we see that $\xi \in G(V)$ corresponds to $\xi' \in G'(V')$. By assumption $V' \times_{\xi', G'} F'$ is representable by a scheme U'. The morphism of schemes $U' \to V'$ corresponding to the projection $V' \times_{\xi', G'} F' \to V'$ is locally of finite presentation by Limits of Spaces, Lemma 049M and Limits, Proposition 01ZC. Hence U' is a locally Noetherian scheme and therefore U' is isomorphic to an object U of $(Noetherian/S)_\tau$. Then U represents $F \times_G V$ as desired.

The implication (2) $\Rightarrow$ (3) is immediate. Assume (3). We will prove (1). Let T be an object of $(Sch/S)_\tau$ and let $T \to G'$ be a morphism. We have to show the functor $F' \times_{G'} T$ is representable by a scheme X over T. Let $\mathcal{B}$ be the set of affine opens of T which map into an affine open of S. This is a basis for the topology of T. Below we will show that for $W \in \mathcal{B}$ the fibre product $F' \times_{G'} W$ is representable

[1] This makes sense even if $\tau \neq fppf$ as the underlying category of $(Sch/S)_\tau$ equals the underlying category of $(Sch/S)_{fppf}$ and the statement doesn't refer to the topology.

by a scheme X_W over W. If $W_1 \subset W_2$ in $\mathcal{B}$, then we obtain an isomorphism $X_{W_1} \to X_{W_2} \times_{W_2} W_1$ because both X_{W_1} and $X_{W_2} \times_{W_2} W_1$ represent the functor $F' \times_{G'} W_1$. These isomorphisms are canonical and satisfy the cocycle condition mentioned in Constructions, Lemma 01LH. Hence we can glue the schemes X_W to a scheme X over T. Compatibility of the glueing maps with the maps $X_W \to F'$ provide us with a map $X \to F'$. The resulting map $X \to F' \times_{G'} T$ is an isomorphism as we may check this locally on T (as source and target of this arrow are sheaves for the Zariski topology).

Let W be an affine scheme which maps into an affine open $U \subset S$. Let $W \to G'$ be a map. Still assuming (3) we have to show that $F' \times_{G'} W$ is representable by a scheme. We may write $W = \lim V_i'$ as a directed limit of affine schemes V_i' of finite presentation over U, see Algebra, Lemma 00QN. Since V_i' is of finite type over an Noetherian scheme, we see that V_i' is a Noetherian scheme. Denote $V_i = V_i'$ but viewed as an object of $(\mathit{Noetherian}/S)_\tau$. As G' is limit preserving can choose an i and a map $V_i' \to G'$ such that $W \to G'$ is the composition $W \to V_i' \to G'$. Since G is the restriction of G' to $(\mathit{Noetherian}/S)_\tau$ the morphism $V_i' \to G'$ is the same thing as a morphism $V_i \to G$ (see above). By assumption (3) the functor $F \times_G V_i$ is representable by an object X_i of $(\mathit{Noetherian}/S)_\tau$. The functor $F \times_G V_i$ is limit preserving as it is the restriction of $F' \times_{G'} V_i'$ and this functor is limit preserving by Limits of Spaces, Lemma 0GDZ, the assumption that F' and G' are limit preserving, and Limits, Remark 05LX which tells us that the functor of points of V_i' is limit preserving. By Lemma 0GE4 we conclude that X_i is locally of finite presentation over S. Denote $X_i' = X_i$ but viewed as an object of $(\mathit{Sch}/S)_\tau$. Then we see that $F' \times_{G'} V_i'$ and the functors of points $h_{X_i'}$ are both extensions of $h_{X_i} : (\mathit{Noetherian}/S)_\tau^{opp} \to \mathit{Sets}$ to limit preserving sheaves on $(\mathit{Sch}/S)_\tau$. By the equivalence of categories of Lemma 0GE3 we deduce that X_i' represents $F' \times_{G'} V_i'$. Then finally

$$F' \times_{G'} W = F' \times_{G'} V_i' \times_{V_i'} W = X_i' \times_{V_i'} W$$

is representable as desired. □